Deep Learning for Minor Motion Magnification: Principles, Applications and Challenges

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fake detection. Across these domains, we identify persistent challenges: limited real-world annotated datasets, sensitivity to illumination and sensor noise, edge distortion, insufficient physical interpretability, weak reproducibility and limited evidence for real-time deployment. Finally, we outline future directions, including physically constrained learning, self-supervised and unsupervised training, multimodal sensing, benchmark construction, uncertainty-aware evaluation and lightweight models for practical systems. By connecting classical principles with current neural approaches, this review aims to provide a coherent taxonomy and a technically grounded roadmap for researchers developing robust and reusable minor motion magnification methods.

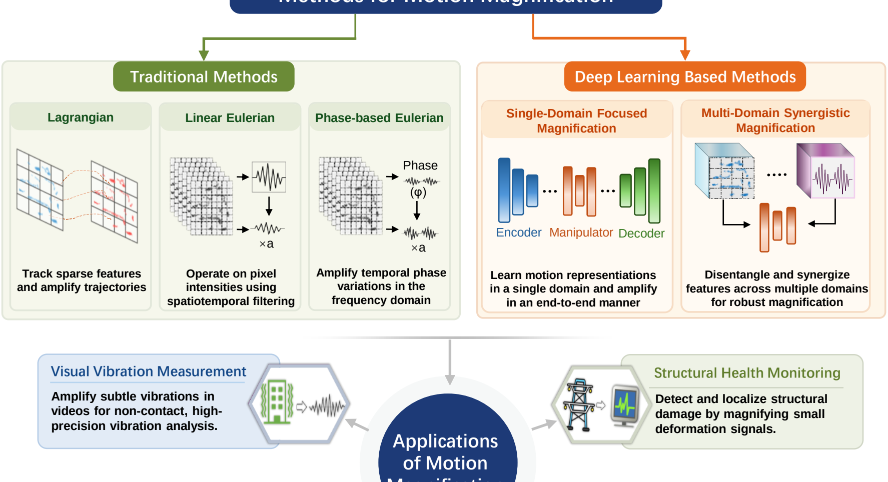
Keywords: Minor motion magnification, Deep learning, Video enhancement, Visual computing, Structural health monitoring

1 Introduction

effective under controlled conditions, they exhibit notable limitations in complex scenarios, such as sensitivity to noise, limited capability in handling non-rigid motion, and restricted amplification factors [23]. In addition, these methods typically require manual parameter tuning, which constrains their adaptability across diverse application environments.

With the rapid advancement of deep learning, data-driven motion magnification methods have attracted increasing attention. By learning motion representations directly from data, these approaches enable end-to-end optimization and improved robustness. Convolutional and transformer-based architectures, for example, have demonstrated strong performance in modeling non-linear motion, suppressing noise, and preserving structural details, thereby extending the applicability of MM to more complex real-world scenarios.

Despite these advances, existing reviews remain limited in several aspects. First, most surveys predominantly focus on classical signal-processing pipelines, with comparatively limited attention given to the systematic organization of deep



Traditional Methods
Lagrangian
Track sparse features and amplify trajectories
Linear Eulerian
×a
Operate on pixel intensities using spatiotemporal filtering
Phase-based Eulerian
Phase
(φ)
×a
Amplify temporal phase variations in the frequency domain
Deep Learning Based Methods
Single-Domain Focused Magnification
Encoder
Manipulator
Decoder
Learn motion representiations in a single domain and amplify in an end-to-end manner
Multi-Domain Synergistic Magnification
Disentangle and synergize features across multiple domains for robust magnification
Visual Vibration Measurement
Amplify subtle vibrations in videos for non-contact, high-precision vibration analysis.
Applications of Motion
Structural Health Monitoring
Detect and localize structural damage by magnifying small deformation signals.

tion" OR "Lagrangian motion magnification" OR "Eulerian motion magnification" OR "phase-based motion magnification"). This query was further complemented with auxiliary terms such as ("deep learning", "vibration", "biomedical", and "micro-expression") to extend coverage toward domain-specific applications.

The overall screening pipeline is illustrated in Figure 2. In the identification stage, a total of 404 records were retrieved from the selected databases. During the initial screening phase, 142 records were excluded based on predefined criteria, including duplicate entries, non-English publications, patents, non-peer-reviewed preprints, and studies not directly related to motion magnification (e.g., general optical flow or large-scale motion tracking).

Subsequently, 114 full-text articles were assessed for eligibility, with evaluation criteria focusing on methodological completeness, relevance to motion magnification, and the availability of quantitative validation. Studies lacking sufficient technical detail, experimental support, or clear methodological contribution were excluded at this stage. Ultimately, 64 representative works were retained for comprehensive analysis cations lacking sufficient technical detail; and (3) duplicate or preliminary versions of already published work.

Representative studies were selected based on a combination of methodological novelty, citation impact, and relevance to key application domains. To further mitigate potential selection bias, results from multiple databases were cross-checked, and preference was given to widely adopted, well-documented, or experimentally validated approaches.

Based on the screened literature, Figure 3 illustrates the development timeline of motion magnification. Since its initial introduction around 2005, the field has undergone steady evolution. Prior to 2018, research was predominantly dominated by handcrafted signal-processing approaches. With the emergence of deep learning-based frameworks, the field has experienced rapid expansion in recent years.

Through this structured investigation, existing motion magnification approaches are categorized into two primary groups: conventional methods and deep learning-based methods. A comparative overview of these methodologies is presented in

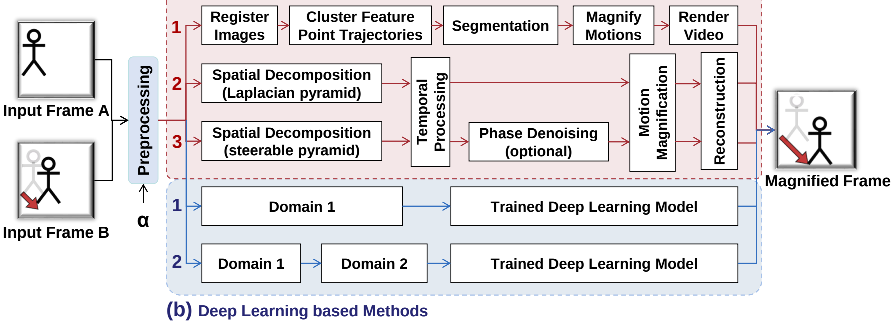


Fig. 4 Motion magnification flowcharts. (a) Traditional methods: (1) Lagrangian, (2) Linear Eulerian, (3) Phase-based Eulerian. (b) Deep learning-based methods: (1) Single-Domain focused, (2) Multi-Domain synergistic.

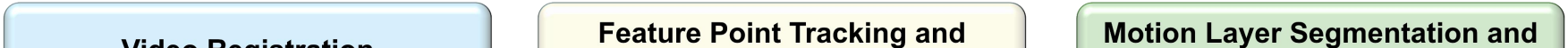


image registration or alignment correction is applied to the input video frames to remove unwanted motion caused by camera shake or slight movements. This process is typically accomplished using techniques like feature point matching[54] and optical flow methods[55–57], which establish pixel correspondences between adjacent frames. This step is essential to stabilize the video, ensuring that subsequent processing focuses solely on the targeted motions, thereby enhancing accuracy and reliability in motion analysis.

II. Feature Point Tracking and Trajectory Clustering

Drawing on the Lagrangian perspective, this approach involves selecting and tracking a series of prominent feature points (such as corners and edges) within the video sequence. These feature points are chosen for their high stability and distinctiveness, ensuring they can be reliably identified across frames. By calculating the displacement of these points in consecutive frames, their motion trajectories are obtained, providing a detailed representation of the movement within the video. This tracking process enables a precise analysis of the dynamic behavior of the observed features.

and noticeable "holes," which are visually disruptive and fall short of human visual standards. To enhance the visual quality, a rendering process is applied. The "holes" can be corrected through techniques like texture synthesis and image interpolation, filling in missing information for a seamless appearance. For the distortion, operations such as noise reduction, contrast enhancement, and color correction are necessary to restore a more natural look, ensuring that the final image is both visually coherent and detailed.

From the above discussion, it is clear that the Lagrangian method is computationally intensive and time-consuming. Its processing steps are tightly interconnected, often requiring multiple manual attempts to select optimal parameters, such as for feature point clustering and motion layer segmentation. Additionally, different scenes necessitate manual parameter adjustments, making it a labor-intensive process to achieve satisfactory magnification results—especially when dealing with long videos or high-resolution data[63–65]. These limitations have hindered the method's further development and widespread application. However, this groundbreaking research has

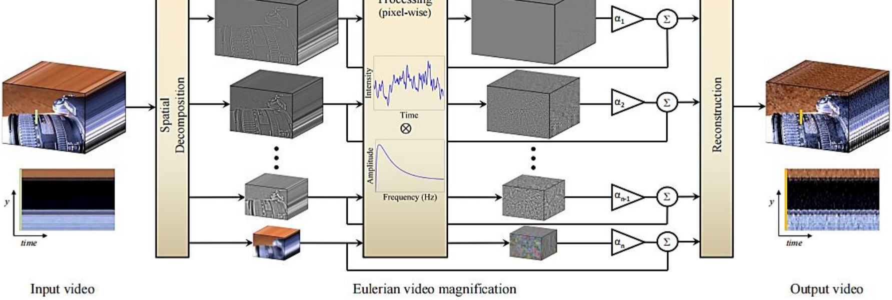


Fig. 6 Overview of the Eulerian video magnification framework[23]. The system employs a three-stage video processing framework: (1) spatial frequency decomposition of the input video, (2) uniform temporal filtering across all frequency bands followed by α-factor amplification, and (3) signal reconstruction through band recombination. This architecture allows flexible parameter optimization, where both temporal filtering characteristics and amplification coefficients can be adaptively configured for specific application scenarios.

by the magnification factor α and added back to the original observed signal $I(x,t)$, yielding the processed intermediate signal:

$$\tilde{I}(x,t) = I(x,t) + \alpha B(x,t) \tag{4}$$

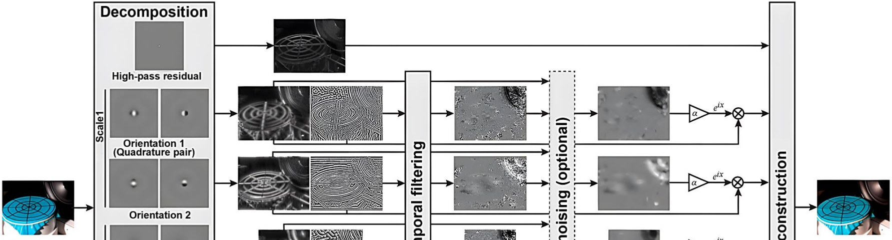
By substituting Equations (2) and (3) into Equation (4), we derive the linearized representation of the synthesized signal:

$$\tilde{I}(x,t) \approx f(x) + (1+\alpha)\delta(t)\frac{\partial f(x)}{\partial x} \tag{5}$$

Under the assumption that the first-order Taylor series expansion remains valid for the augmented displacement, Equation (5) can be mapped back into the spatial domain via an inverse Taylor approximation:

$$\tilde{I}(x,t) \approx f(x + (1+\alpha)\delta(t)) \tag{6}$$

each coefficient is characterized by a local amplitude and a local phase. A temporal band-pass filter is subsequently applied to the time-varying phase of each sub-band to isolate the desired motion frequencies. The isolated phase perturbation is then amplified by a specified factor and linearly recombined with the original phase profile. Consequently, in the final synthesized video, the underlying motion-encoded phase is magnified while the local amplitude remains unaltered, as illustrated in Figure 7.



within a specific temporal frequency band, a DC-balanced temporal band-pass filter is applied to the time-varying phase $\omega(x + \delta(t))$. Assuming the motion spectrum lies entirely within the filter's passband, the static spatial component ωx is effectively suppressed, yielding the filtered phase signal:

$$B_\omega(x, t) = \omega\delta(t) \tag{9}$$

To construct the motion-magnified sub-band, the band-passed phase $B_\omega(x, t)$ is scaled by the magnification factor α and added to the phase of the original sub-band $S_\omega(x, t)$:

$$\hat{S}_\omega(x, t) = S_\omega(x, t)e^{i\alpha B_\omega(x,t)} = A_\omega e^{i\omega(x+(1+\alpha)\delta(t))} \tag{10}$$

The effective displacement within the modified sub-band $\hat{S}_\omega(x, t)$ is now linearly scaled to $(1 + \alpha)\delta(t)$. Finally, summing the modified components across all spatial frequencies reconstructs the motion-magnified sequence $\hat{I}(x, t) = f(x + (1 + \alpha)\delta(t))$, thereby completing the inverse pyramid synthesis.

tude) constitutes the core signal-processing mechanism behind PEVM's exceptional robustness.

Despite the superior noise resilience and visual quality offered by PEVM, its practical execution involves substantial computational trade-offs when contrasted with LEVM and Lagrangian alternatives. Mathematically, LEVM achieves a strict linear time and space complexity of $\mathcal{O}(N)$, where N denotes the total number of pixels within a frame. This linear scaling yields extreme computational efficiency, rendering real-time execution on standard CPU architectures highly feasible. In contrast, the complex steerable pyramid decomposition utilized in PEVM demands $\mathcal{O}(K \cdot L \cdot N)$ space to store complex coefficients across L scales and K orientations, while its temporal complexity—primarily driven by multi-scale Fast Fourier Transforms (FFTs)—reaches $\mathcal{O}(K \cdot L \cdot N \log N)$. This heightened memory footprint and computational overhead create a notable bottleneck for high-resolution video streams, typically necessitating parallelized GPU acceleration to unlock real-time processing.

Limitations notwithstanding, these Eulerian operational profiles remain signifi-

Table 1 Taxonomy, Characteristics, and Functional Roles of Representative MM Datasets.

Dataset	**Data Type**	**Ground Truth (GT)**	**Primary Role**	**Typical Scenario / Key Characteristics**
Oh et al. [33]	Synthetic	Pixel-level motion field	Training & Testing	Multi-object non-rigid 2D warping; 200k backgrounds / 7k foregrounds.
AxialMM Dataset [78]	Synthetic	Directional displacement field	Training & Testing	Directionally decoupled harmonic motion along specified X/Y axes.
DIS5K-StickPNG [40, 41]	Synthetic	Multi-level synthetic annotations	Robustness Test	High-resolution background textures with injected Poisson noise and Gaussian blur.
Real-world Benchmark Dataset [23, 32, 33]	Real-world	None	Validation & Demo	Consolidated standard suite: *Static Mode* (Baby, Guitar, Crane) and *Dynamic Mode* (Face, Swing, Gunshot).

(a)

(b)

(c)

Fig. 8 Example frames from dataset. (a) Pasted foreground objects using segmentation from PASCAL VOC and background from MS COCO dataset. (b) Only has background moving to ensure

Table 2 Systematic Categorization of Evaluation Indicators and Targeted Physical Dimensions.

Targeted Dimension	Primary Indicators	Reference Mode	Physical Assessment Goal
Spatial Fidelity & Clarity	MSE [82], PSNR [82], SSIM [83], LPIPS [84]	Full-Reference	Pixel reconstruction accuracy and structural/perceptual alignment.
Motion Amplitude Accuracy	Temporal slice track (X-T/Y-T), Error Margin	Full / Implicit Ref	Quantifies if output displacement matches the targeted scale ($\alpha \cdot \delta$).
Temporal Consistency	Temporal wrap loss, Inter-frame variance	Full-Reference	Detects flickering artifacts, jittering, or non-uniform frame propagation.
Frequency Fidelity	Power Spectral Density (PSD), Fourier Error	Implicit Reference	Verifies if the amplified frequency matches the source vibration spectrum.
Environmental Perturbations	MUSIQ [85], HyperIQA [86], MANIQA	No-Reference	Evaluates sensitivity to photon noise, shadows, and illumination changes.

features extracted by CNNs trained on the Berkeley-Adobe Perceptual Patch Similarity (BAPPS) dataset, capturing subtle differences and reflecting human perceptual characteristics under high magnification factors.

(b) Motion Amplitude Accuracy: This indicator quantifies whether the actual physical displacement of an object in the magnified output closely matches the targeted mathematical scaling factor ($\alpha \cdot \delta$). On synthetic configurations, this is frequently computed by evaluating the error margin against the explicit ground truth warping field. On real-world sequences lacking explicit fields, researchers typically utilize sub-pixel tracking along temporal spatial-temporal slices (X-T or Y-T plots) or cross-reference the output displacement with the physical measurements of available laser displacement sensors [78].

(c) Temporal Consistency: Temporal consistency ensures that the magnified video sequence is smooth and free from inter-frame flickering artifacts or non-uniform artificial acceleration. Handcrafted Eulerian networks achieve this via strict temporal bandpass filtering, whereas deep neural networks optimize this dimension by integratboundary artifacts, while specialized lightweight architectures explicitly incorporate parameter-efficient scaling or differentiable architecture searches to minimize inference delay for practical deployment.

2.3.2 Single-Domain Focused Magnification

Traditional motion magnification methods rely on manually designed features and filters, which often exhibit limited robustness in complex or non-rigid motion scenarios and are sensitive to noise and illumination variations. With the rapid advancement of computational hardware and deep learning techniques, data-driven approaches have emerged, enabling more accurate and stable motion amplification through end-to-end learning frameworks.

To provide a structured and interpretable overview of deep learning-based methods, we categorize existing approaches into two representative paradigms: Single-Domain Focused Magnification (SDFM) and Multi-Domain Synergistic Magnification amplification, it is classified as **MDSM** (e.g., MD-VMM [37]).

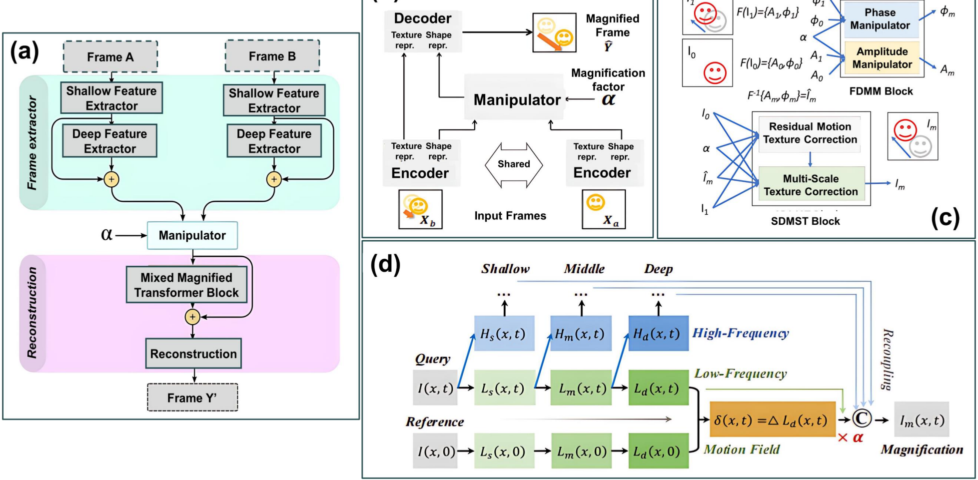
- **Rule 3 (Multi-Modal Integration):** If the architecture incorporates heterogeneous sensing modalities (e.g., combining RGB frames with event-based signals) and performs joint motion reasoning across modalities, it is classified as **MDSM** (e.g., EB-VMM [42]).

To provide a representative blueprint of the current literature, Table 3 presents a structured technical taxonomy and technical comparison mapping prominent deep learning methods across both paradigms based on their core architectural characteristics. For conceptual and architectural clarity, Fig. 9 highlights the fundamental distinction between SDFM and MDSM in terms of representation space and amplification mechanism, whereas Fig. 10 provides a structural comparison of representative network architectures across the two paradigms.

Table 3 Refined taxonomy and hardware-software configuration matrix of prominent lear

structed by applying random local pixel displacements, affine transformations, or 2D sinusoidal vibrations to high-quality static images to establish clean ground-truth reference frames with specified magnification factors (α). LN-VMM [38] improves on this by introducing high-throughput training via lightweight surrogate models. For MDSM, MD-VMM [37] and FD-VMM [41] employ frequency-specialized synthetic sets containing decoupled low-frequency spatial structures and high-frequency motion details. Conversely, EB-VMM [42] represents a unique self-supervised paradigm trained on the multi-modal Ev-Magnify dataset, utilizing event-reconstruction consistency (L_{rc}) instead of explicit frame-level ground-truths.

In terms of quantitative assessment and computational complexity, PSNR and SSIM serve as the universal standard evaluation metrics across all seven frameworks to benchmark structural fidelity and noise resistance. Additional metrics include the axial motion score for LBA-VMM [35], temporal consistency indices for STB-VMM [36], and phase-SNR for MD-VMM [37]. Computationally, the models diverge significantly: LN-VMM [38] and LB-VMM [33] deliver real-time processing capabilities with



(a)
Frame A
Frame B
Frame extractor
Shallow Feature Extractor
Deep Feature Extractor
α
Manipulator
Reconstruction
Mixed Magnified Transformer Block
Reconstruction
Frame Y'
Decoder
Texture repr.
Shape repr.
Magnified Frame
Ŷ
Magnification factor
α
Manipulator
Encoder
Shared
Input Frames
X_b
X_a
F(I_1)={A_1,ϕ_1}
F(I_0)={A_0,ϕ_0}
Phase Manipulator
Amplitude Manipulator
ϕ_m
A_m
FDMM Block
F^{-1}{A_m,ϕ_m}=Î_m
Residual Motion Texture Correction
Multi-Scale Texture Correction
SDMST Block
(c)
(d)
Shallow
Middle
Deep
H_s(x,t)
H_m(x,t)
H_d(x,t)
High-Frequency
Query
I(x,t)
L_s(x,t)
L_m(x,t)
L_d(x,t)
Low-Frequency
Recoupling
Reference
I(x,0)
L_s(x,0)
L_m(x,0)
L_d(x,0)
δ(x,t) =△ L_d(x,t)
× α
Motion Field
I_m(x,t)
Magnification

scenarios, the authors incorporate localized non-linear mapping operations to suppress distortion, formulated as:

$$G_m(M_a, M_b, \alpha) = M_a + h(\alpha \cdot g(M_b - M_a)), \tag{17}$$

where $g(\cdot)$ and $h(\cdot)$ denote standard convolutional steps and non-linear activation functions, respectively.

During the network training phase, optimization is guided strictly by an L_1 reconstruction loss penalizing the divergence between the predicted output $\hat{Y}$ and the ground-truth magnified frame Y. Empirical evaluations by the authors demonstrated that integrating more sophisticated objectives, such as perceptual loss [89] or adversarial loss [90], yielded negligible performance improvements. From an optimization perspective, the constant gradient magnitude of the L_1 norm ensures robust, linear convergence and mitigates gradient explosion risks. Conversely, the highly non-convex loss surfaces characteristic of adversarial and perceptual objectives tend to destabitheir architecture effectively isolates and magnifies subtle structural displacements along user-specified coordinate axes, drastically improving motion legibility in complex mechanical systems. Mechanistically, LBA-VMM introduces directional spatial filtering priors to decouple orthogonal noise. While this directional assumption preserves highly sharp boundaries along specified axes, the network sacrifices real-time throughput ($\sim$15 FPS) due to the multi-stage directional projections. Its ultimate failure mode occurs when processing chaotic, non-rigid structural fields, where the rigid linear directional assumption breaks down completely and yields severe geometric ghosting artifacts.

Addressing the high computational footprint required for deployment on edge devices, Singh et al. [38] developed a lightweight, real-time alternative (LN-VMM) utilizing an aggressive feature-sharing encoder configuration. By pairing an appearance preservation stream with an explicit edge-preservation loss (L_e), their framework scales down parameter overhead significantly to achieve outstanding real-time edge processing ($\sim$90 FPS). The foundational assumption of LN-VMM is that spatial tional efficiency and noise handling. This lightweight design operates 2.7 times faster with 4.2 times fewer FLOPs than the baseline, enabling real-time 1080p inference without sacrificing generation quality.

Despite these rapid architectural variations, the single-Domain paradigm retains inherent structural constraints. The clear advantage of SDFM frameworks lies in their implementation simplicity and low computational latency, rendering them highly effective for isolated, well-defined 2D spatial displacements. However, because these methods operate monolithically within homogeneous coordinate spaces, they lack the capacity for explicit frequency or multimodal feature decoupling. Consequently, when deployed in complex real-world scenes characterized by low signal-to-noise ratios (SNR), wideband environmental interference, or non-rigid structural deformations, SDFM models frequently suffer from severe texture blurring, geometric distortion, and noise amplification, marking a boundary that necessitates the transition to Multi-Domain architectures.

such as temporal inconsistency or spectral leakage.

Beyond mathematical domains, MDSM also extends to multi-modal sensing (Rule 3). EB-VMM [42], for instance, combines RGB frames with event-based data to exploit the high temporal resolution of event cameras. This design enables more accurate capture of subtle temporal variations and alleviates limitations of frame-based sampling.

However, event-based methods depend on sensor-specific characteristics, such as contrast thresholds, which may lead to missing information in low-motion or static regions. In addition, cross-modal alignment introduces additional computational complexity.

Other approaches, such as σ-θ Net [94], attempt to achieve domain disentanglement within a learned representation space without explicit reliance on external modalities. By separating scale and directional components, these methods provide a lightweight alternative, although their performance may degrade in highly irregular motion scenarios.

ing for analytical formulations. SDFM approaches typically operate within a single representation space with implicit disentanglement, while MDSM methods introduce explicit transformations or multi-domain fusion to separate motion and appearance components.

Despite these advantages, deep learning models introduce new trade-offs. The lack of explicit physical constraints may reduce interpretability and lead to data-dependent behavior. Common failure cases include temporal flickering in frame-wise models, artifacts introduced by frequency-domain processing, and sensitivity to modality alignment in multi-modal systems. Furthermore, their computational demands can be substantial, posing challenges for real-time or resource-constrained deployment without model optimization.

Consequently, the selection of an appropriate motion magnification paradigm is task-dependent and constrained by the operational setting:

- **Eulerian Methods** are well-suited for resource-constrained environments and peri-

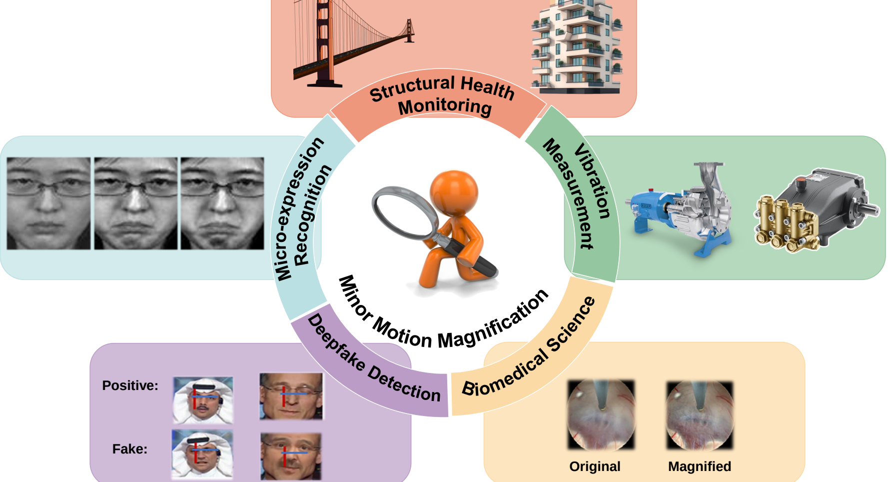

Structural Health Monitoring
Vibration Measurement
Biomedical Science
Deepfake Detection
Micro-expression Recognition
Minor Motion Magnification
Positive:
Fake:
Original
Magnified

Table 4 Empirical mapping of typical technical profiles across prominent motion magnification application domains.

Domain	Typical Input Data	Target Scale	Validation Method	Primary Indicators	Common Constraints
Visual Vibration Measurement	High-speed RGB/-Mono ($>$ 250 fps); Macro lens	Sub-pixel to sub-mm (10^{-3} to 10^{-1} px)	Laser Doppler Vibrometer comparison	PSD peaks, FRF, Fourier Error	High-frequency jitter; specular reflections
Structural Health Monitoring	Standard 4K/HD RGB (30/60 fps); UAV/Telephoto	Sub-mm to mm structural drift	Accelerometers or strain gauges	ODS, MAC	Wind-induced camera motion; atmospheric shimmer
Biomedical Analysis	RGB (30 fps); Near-Infrared (NIR)	Micro-vessel pulsing; sub-px motion	PPG or ECG reference	BPM error, RR, SNR	Motion artifacts; illumination variability
Micro-Expression Recognition	Facial video (60–200 fps)	Sub-pixel facial muscle motion	FACS annotations	Accuracy, F1-score	Head motion; occlusion

on handcrafted or learned features (e.g., LBP-TOP) [106, 127, 128].

A key challenge is the coexistence of micro-expressions with larger head motion or speech-related movement. Without effective motion disentanglement, magnification may introduce artifacts such as edge distortion or texture inconsistency [129]. Parameter tuning, including the magnification factor α, remains application-dependent [130, 131].

3.5 Deepfake Detection

Deepfake detection leverages MM to reveal subtle temporal inconsistencies in synthesized media, including abnormal motion patterns or missing physiological cues [132–135]. Input data typically consist of compressed video streams (e.g., H.264 or AV1), often with varying quality and bitrate.

Evaluation is conducted through cross-dataset testing (e.g., FaceForensics++) using metrics such as AUC-ROC, Equal Error Rate (EER), and classification accuracy ducing mass?stiffness constraints can restrict magnification to motion patterns that satisfy realistic oscillation behavior, thereby suppressing stochastic noise.

In addition, improving model interpretability is essential for safety-critical applications, such as biomedical monitoring or precision industrial diagnostics. Future work should explore self-explainable mechanisms, including frequency-aware attention maps or phase-consistency fields, to enable practitioners to verify whether amplified signals correspond to genuine physical phenomena rather than algorithmic artifacts.

4.2 Domain Adaptation and Generative Minor Motion Synthesis

Another major limitation arises from the gap between synthetic training data and real-world deployment environments. Due to the difficulty of acquiring ground-truth minor motion in natural settings, most existing methods rely on synthetic datasets generated through simplified transformations, such as linear 2D warping of static improved efficiency.

In addition, mixed-precision quantization tailored for sub-pixel motion representation can reduce computational cost while preserving numerical stability. Achieving an effective balance between efficiency and accuracy is critical for enabling real-time MM systems on edge devices, such as portable inspection platforms or embedded structural monitoring systems.

5 Conclusion

This review presents a structured and comprehensive synthesis of *minor motion magnification* (MM, also referred to as video motion magnification, MM), with the primary objective of clarifying its theoretical foundations, methodological evolution, and practical deployment characteristics. Specifically, this work establishes a unified mathematical formulation that abstracts diverse approaches under a common reconstruction objective, and proposes a two-tier taxonomy that categorizes

Although this work is a review article and does not introduce new experimental models or datasets, we emphasize reproducibility and transparency in the literature screening and taxonomy construction process.

The literature search strategy, including database sources, search date, and query formulation, as well as the inclusion and exclusion criteria, are explicitly described in Section 1. These details enable other researchers to replicate the retrieval procedure under similar conditions.

The representative studies analyzed in this review are reflected in the reference list and are discussed throughout the manuscript, ensuring traceability of the reviewed evidence.

Due to copyright restrictions and database access limitations, the complete set of initially retrieved records and intermediate screening logs cannot be publicly redistributed. However, the methodological transparency provided in this study supports reproducibility at the procedural level.

sion, Project administration, Resources. **Bo Peng:** Investigation, Formal analysis. **Ningsheng Liao:** Supervision, Writing-Review & Editing.

References

[1] Collins, R.: On the microfoundations of macrosociology. American journal of sociology **86**(5), 984–1014 (1981)

[2] Gibson, C.C., Ostrom, E., Ahn, T.-K.: The concept of scale and the human dimensions of global change: a survey. Ecological economics **32**(2), 217–239 (2000)

[3] Al-Naji, A., Lee, S.-H., Chahl, J.: An efficient motion magnification system for real-time applications. Machine Vision and Applications **29**, 585–600 (2018)

b-mode visualization of magnetomotive ultrasound imaging. Available at SSRN 5134541 (2022)

[13] Terem, I., *et al.*: Revealing sub-voxel motions of brain tissue using phase-based amplified mri (amri). Magnetic resonance in medicine **80**(6), 2549–2559 (2018)

[14] Arul Prakash, S.K., *et al.*: Detection of system compromise in additive manufacturing using video motion magnification. Journal of Mechanical Design **142**(3), 031109 (2020)

[15] Taslicay, C., et al.: Visualization and identification of nonlinear structural dynamics via phase-based motion magnification. Available at SSRN 5165980 (2022)

[16] Civera, M., Zanotti Fragonara, L., Surace, C.: An experimental study of the feasibility of phase-based video magnification for damage detection and localisation

24(3), 519–526 (2005)

[26] Wadhwa, N., *et al.*: Phase-based video motion processing. ACM Transactions on Graphics (TOG) **32**(4), 1–10 (2013)

[27] Wadhwa, N., *et al.*: Riesz pyramids for fast phase-based video magnification. In: 2014 IEEE International Conference on Computational Photography (ICCP), pp. 1–10 (2014). IEEE

[28] Liu, L., *et al.*: Enhanced eulerian video magnification. In: 2014 7th International Congress on Image and Signal Processing, pp. 1–6 (2014). IEEE

[29] Elgharib, M., *et al.*: Video magnification in presence of large motions. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 1–10 (2015). IEEE

fication. In: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, pp. 1–10 (2023). IEEE

[39] Chen, L., Peng, C., Zhao, B.: Novel multi-task learning for motion magnification. IEEE Transactions on Circuits and Systems for Video Technology **33**(10), 5973–5985 (2023)

[40] Wang, F., *et al.*: Eulermormer: Robust eulerian motion magnification via dynamic filtering within transformer. In: Proceedings of the AAAI Conference on Artificial Intelligence, pp. 1–10 (2024). AAAI

[41] Wang, F., *et al.*: Frequency decoupling for motion magnification via multi-level isomorphic architecture. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 1–10 (2024). IEEE

puter Vision, Prague, Czech Republic, May 11-14, 2004. Proceedings, Part IV 8, pp. 1–10 (2004). Springer

[51] Ahmed, A.M., et al.: An overview of eulerian video motion magnification methods. Computers & Graphics (2023)

[52] Yeung, P.K.: Lagrangian investigations of turbulence. Annual review of fluid mechanics **34**(1), 115–142 (2002)

[53] Bennett, A.: Lagrangian Fluid Dynamics. Cambridge Monographs on Mechanics. Cambridge University Press, Cambridge (2006)

[54] Mount, D.M., Netanyahu, N.S., Le Moigne, J.: Efficient algorithms for robust feature matching. Pattern recognition **32**(1), 17–38 (1999)

[55] Gibson, J.J.: The Perception of the Visual World. Houghton Mifflin, Boston

October 11–14, 2016, Proceedings, Part IV 14, pp. 1–10 (2016). Springer

[65] Revaud, J., *et al.*: Epicflow: Edge-preserving interpolation of correspondences for optical flow. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 1–10 (2015). IEEE

[66] Lucas, B.D., Kanade, T.: An iterative image registration technique with an application to stereo vision. In: IJCAI'81: 7th International Joint Conference on Artificial Intelligence, pp. 1–10 (1981). Morgan Kaufmann

[67] Burt, P.J., Adelson, E.H.: The laplacian pyramid as a compact image code. In: Readings in Computer Vision, pp. 671–679. Morgan Kaufmann, San Francisco (1987)

[68] Adelson, E.H., *et al.*: Pyramid methods in image processing. RCA engineer **29**(6), 33–41 (1984)

axial video motion magnification. In: Leonardis, A., Ricci, E., Roth, S., Russakovsky, O., Sattler, T., Varol, G. (eds.) Computer Vision – ECCV 2024, pp. 179–195. Springer, Cham (2025)

[79] Lin, T.-Y., *et al.*: Microsoft coco: Common objects in context. In: Computer Vision–ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6-12, 2014, Proceedings, Part V 13, pp. 1–10 (2014). Springer

[80] Everingham, M., *et al.*: The pascal visual object classes (voc) challenge. International journal of computer vision **88**, 303–338 (2010)

[81] Pan, Z., Geng, D., Owens, A.: Self-supervised motion magnification by backpropagating through optical flow. In: Oh, A., Naumann, T., Globerson, A., Saenko, K., Hardt, M., Levine, S. (eds.) Advances in Neural Information Processing Systems, vol. 36, pp. 253–273. Curran Associates, Inc., ??? (2023)

works. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 1–10 (2017). IEEE

[91] Vaswani, A., et al.: Attention is all you need. Advances in neural information processing systems **30** (2017)

[92] Liu, Z., *et al.*: Swin transformer: Hierarchical vision transformer using shifted windows. In: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 1–10 (2021). IEEE

[93] Ha, H., Hyun-Bin, O., Jun-Seong, K., Kwon, B.-K., Sung-Bin, K., Tran, L.-T., Kim, J., Bae, S.-H., Oh, T.-H.: Revisiting learning-based video motion magnification for real-time processing. ArXiv **abs/2403.01898** (2024)

[94] Singh, J., Kumar Vipparthi, S., Murala, S., Kosuru, G.S.R., Almarzouqi, H.: Learnable directional scale space filters for video motion magnification

(2021)

[103] Bharadwaj, S., *et al.*: Computationally efficient face spoofing detection with motion magnification. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops, pp. 1–10 (2013). IEEE

[104] Das, R., Negi, G., Smeaton, A.F.: Detecting deepfake videos using euler video magnification. arXiv preprint arXiv:2101.11563 (2021)

[105] Mehra, A., *et al.*: Motion magnified 3-d residual-in-dense network for deepfake detection. IEEE Transactions on Biometrics, Behavior, and Identity Science **5**(1), 39–52 (2022)

[106] Le Ngo, A.C., *et al.*: Eulerian emotion magnification for subtle expression recognition. In: 2016 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 1–10 (2016). IEEE

synthetic-to-real image correlation for instance recognition in structure monitoring. The Visual Computer **41**(1), 281–301 (2025)

[116] Cha, Y.-J., Chen, J.G., Büyüköztürk, O.: Motion magnification based damage detection using high speed video. In: Structural Health Monitoring 2015, pp. 1–10 (2015)

[117] Sarrafi, A., Mao, Z., Niezrecki, C., Poozesh, P.: Vibration-based damage detection in wind turbine blades using phase-based motion estimation and motion magnification. Journal of Sound and Vibration **421**, 300–318 (2018)

[118] Figueiredo, E., et al.: Structural health monitoring algorithm comparisons using standard data sets (2009)

[119] Teed, Z., Deng, J.: Raft: Recurrent all-pairs field transforms for optical flow. In: Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK,

motion magnification and pre-trained neural network. In: 2021 IEEE International Conference on Image Processing (ICIP), pp. 549–553 (2021)

[130] Wang, Y.-J., *et al.*: Effective recognition of facial micro-expressions with video motion magnification. Multimedia Tools and Applications **76**, 21665–21690 (2017)

[131] Le Ngo, A.C., *et al.*: Micro-expression motion magnification: Global lagrangian vs. local eulerian approaches. In: 2018 13th IEEE International Conference on Automatic Face & Gesture Recognition (FG 2018), pp. 1–10 (2018). IEEE

[132] Verdoliva, L.: Media forensics and deepfakes: an overview. IEEE journal of selected topics in signal processing **14**(5), 910–932 (2020)

[133] Mirsky, Y., Lee, W.: The creation and detection of deepfakes: A survey. ACM computing surveys (CSUR) **54**(1), 1–41 (2021)